



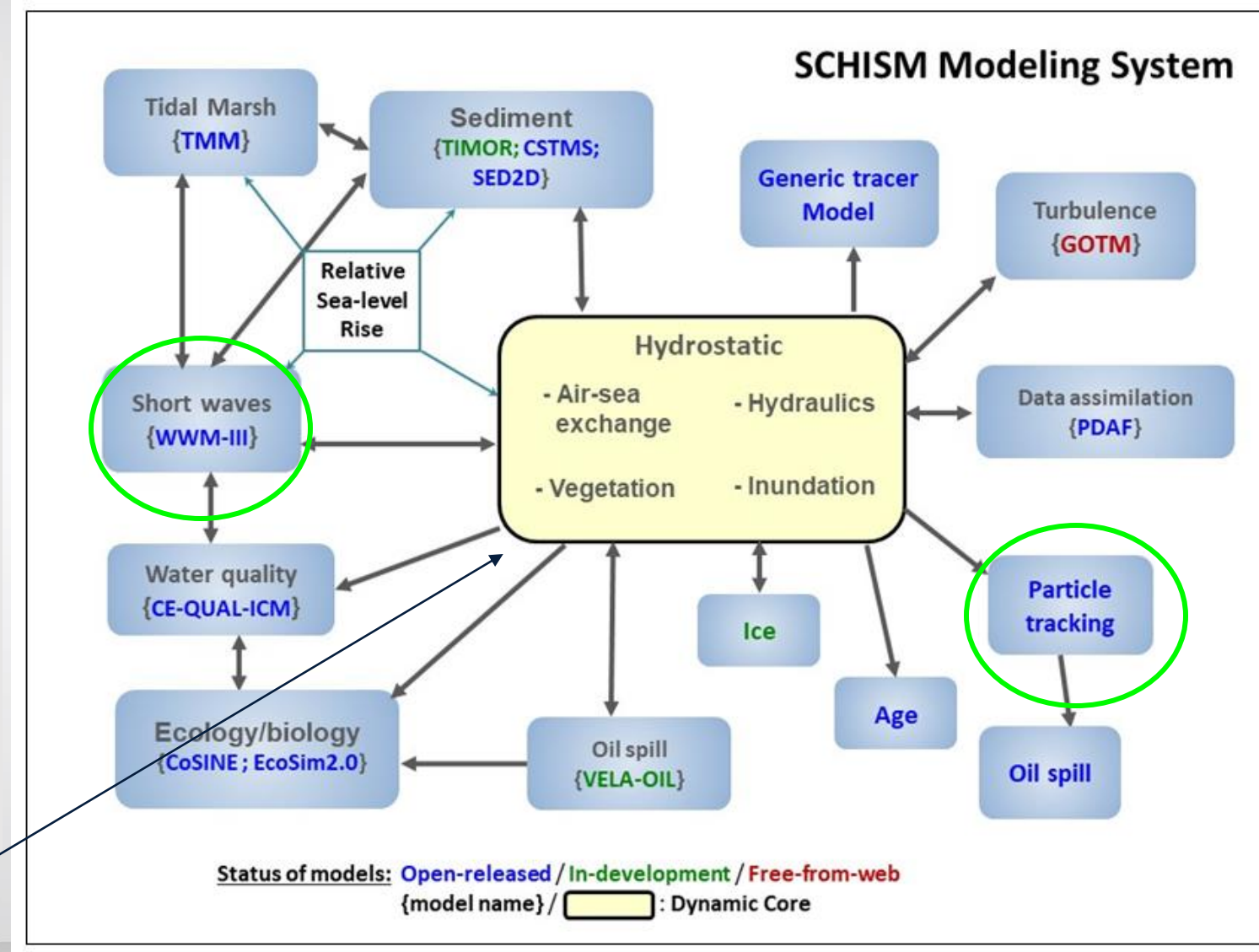
Hydrologic Connectivity Between the Everglades and the Florida Bay Through Coastal Mangroves and Freshwater Sloughs

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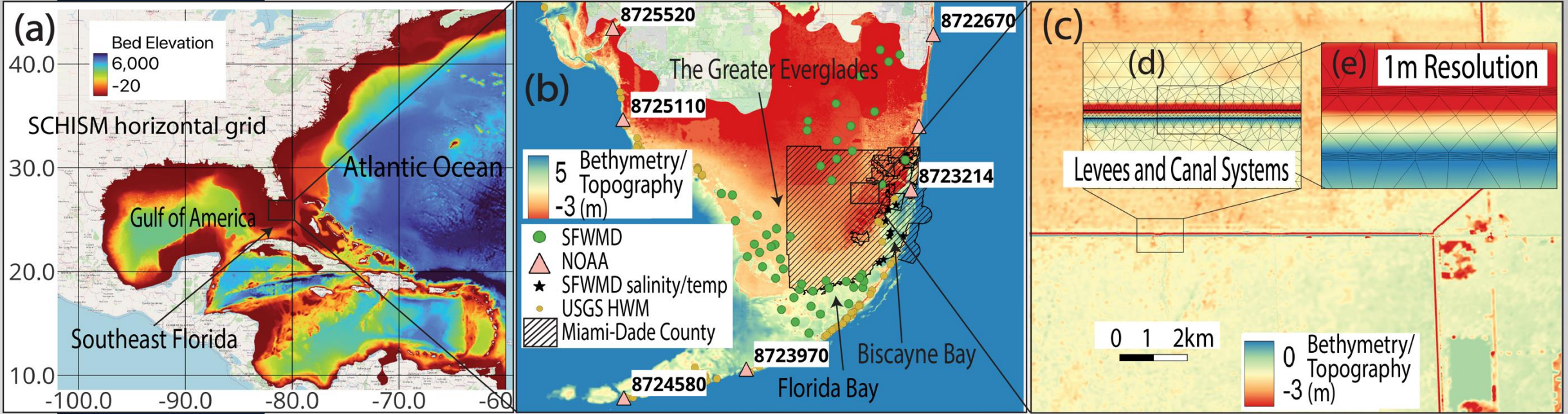
Hydrodynamic modeling with a Semi-implicit Cross-scale Hydrosience Integrated System Model (SCHISM)

- The new generation of unstructured grid, open-source model, SCHISM is the core model of the NOAA's operational forecast systems (<https://polar.ncep.noaa.gov/estofs/>) and being integrated into the NOAA's CoastalApp (<https://github.com/noaa-ocs-modeling/CoastalApp>), is selected by EPA's Chesapeake Bay Program and Texas Water Development Board's coastal operational forecast (<https://midgewater.twdb.texas.gov/index.html>).
- Feature 1:** seamless simulation on a single mesh for multi-scale processes with low computational costs.
- Feature 2:** the model grid can faithfully follow the topography and bathymetry without any smoothing.

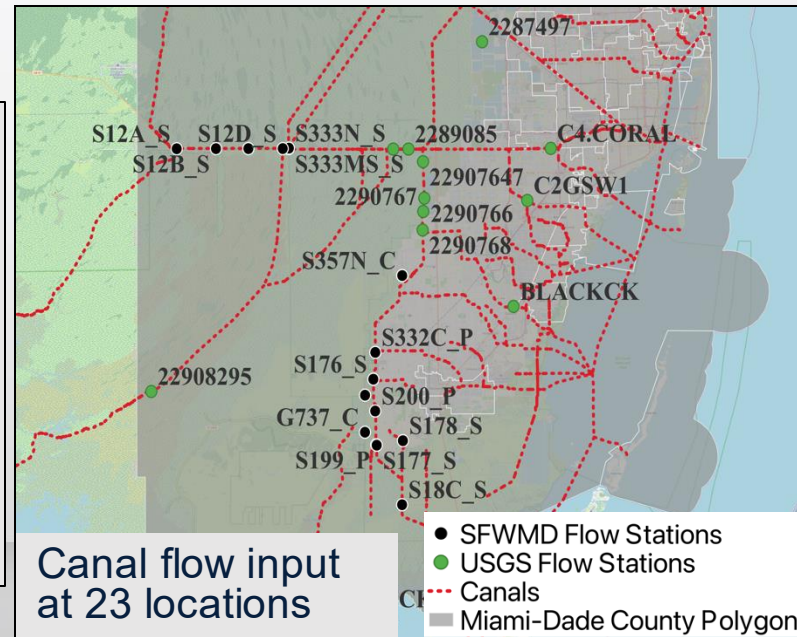
Coupled with National Water Model's river discharge (Huang et al., 2021)



Model's horizontal grid using flexible resolution for cross scale systems Worlds Ahead



- Model grid has higher resolution for watershed, and coarser resolution for the open ocean.
- The grid is refined based on the selection of the focus area.
- Forcing: tides, waves, HYCOM, atmospheric forcing, freshwater discharge.



Simulation period:
2024-1-1 to 2024-12-31
Simulated variables:
Surface water elevation, flow
velocity, salinity, temperature, and
fluxes through transects

Water budget through Taylor River Slough

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(a)

S199_P
S177_S

Transect 2

Transect 1

(b)

Condition 1

Condition 2

Flow through Transect 1

Flow through Transect 2

Flow through S177_S+S199_P

Days since 2024-1-1

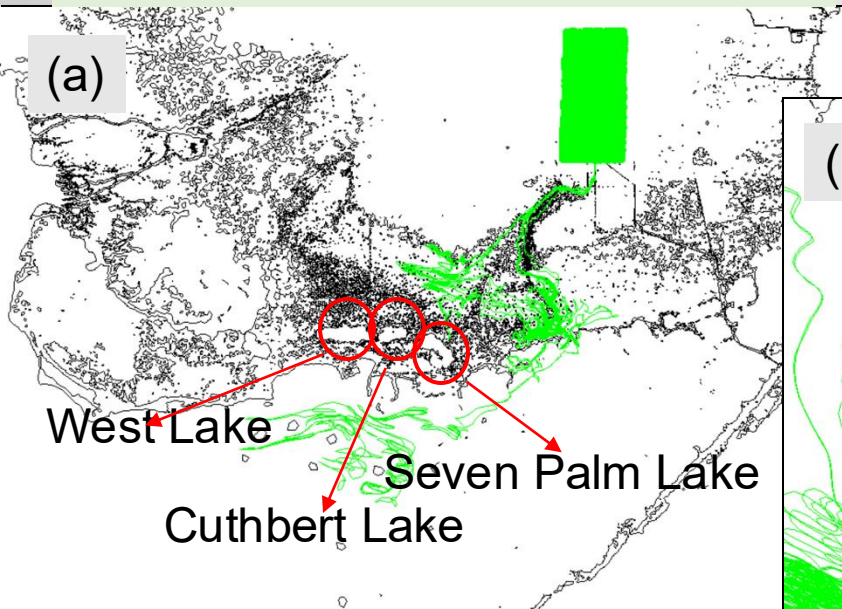
Condition 1: Water overflowed Canals
Transect 2 – Transect 1 > 0 Water overflowed TRS
Canal flow – Transect 1 or Transect 2 > 0

Condition 2: Saltwater intrusion from Florida Bay
Canal flow – Transect 1 or Transect 2 < 0
Transect 2 – Transect 1 < 0

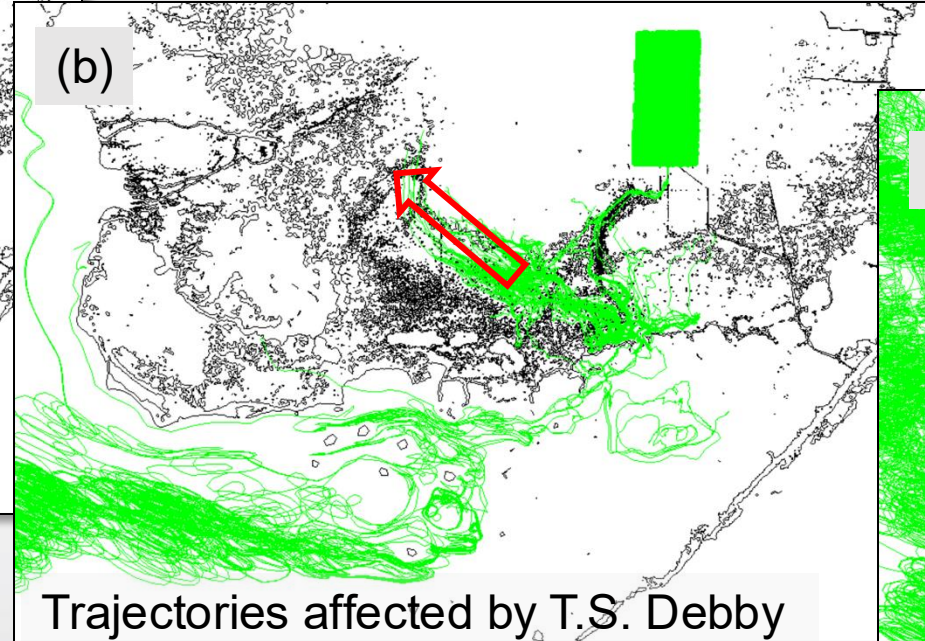
Surface water movement by Lagrangian Particle Tracking

More water particles Released on Jun 11

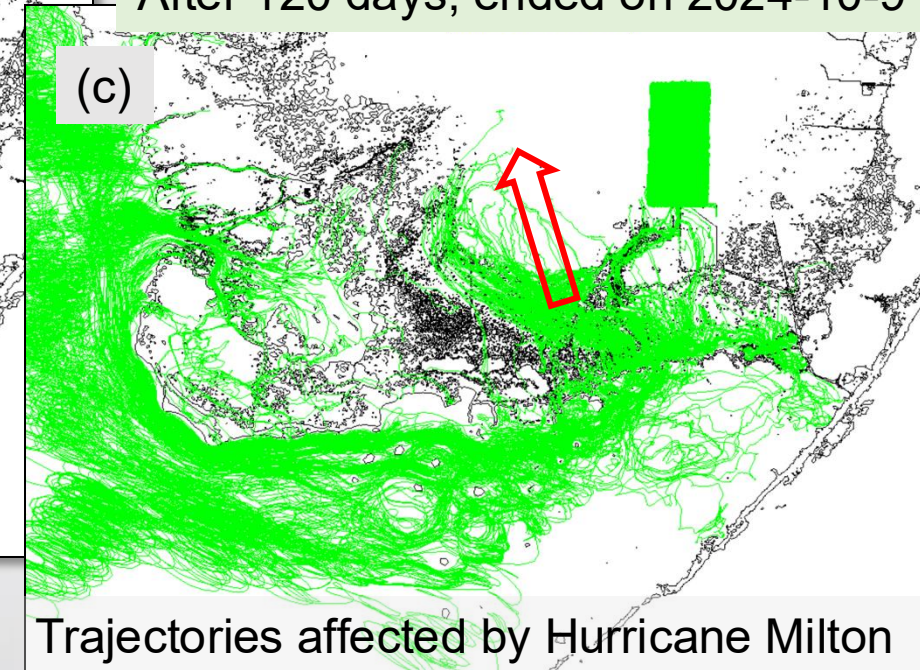
After 30 days, ended on 2024-7-11



After 56 days, ended on 2024-8-6



After 120 days, ended on 2024-10-9



FIU

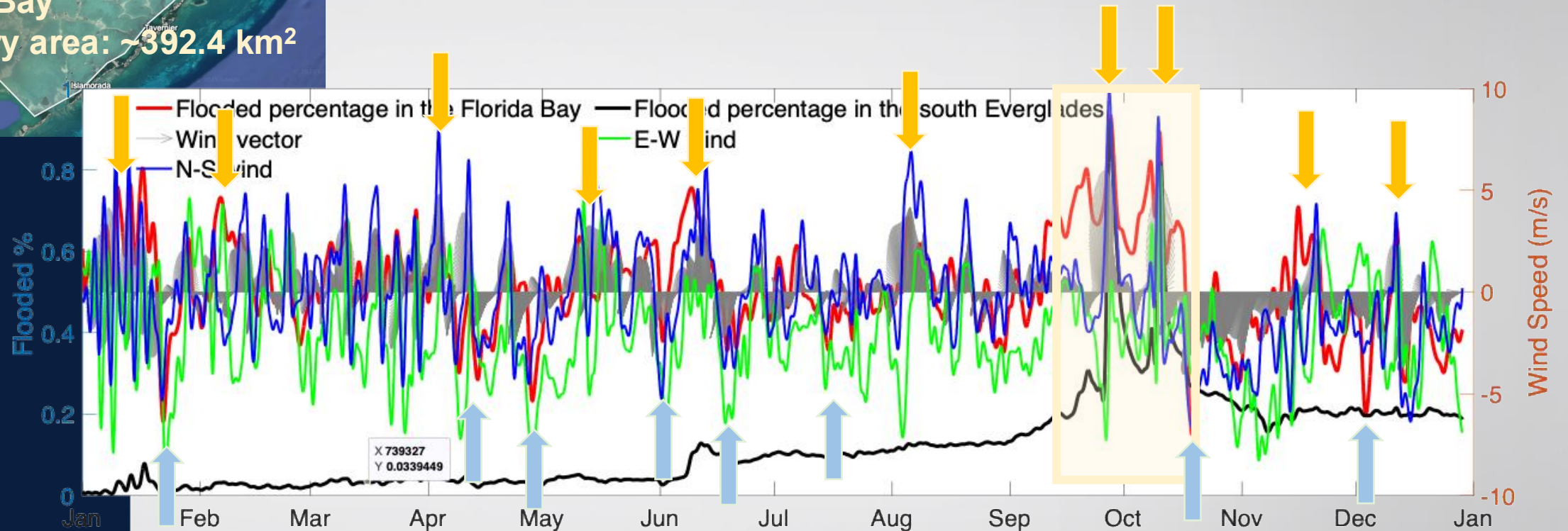
- It took ~30 days for particles to reach Seven Palm Lake
- Seven Palm Lake received particles mainly through Taylor Slough
- West Lake and Cuthbert Lake received particles from those transported to the Florida Bay.

Wind induced flooding extent

Southern Everglades
Initial dry area: ~725.6 km²

Florida Bay
Initial dry area: ~392.4 km²

Southerly and Westerly wind --- more flooded areas
Northerly and Easterly wind --- less flooded areas



	Wind	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Everglades	E-W	0.29	0	0.14	0.3	-0.22	0.43	0.2	0.36	0.05	0.7	0.49	0.62
725.6 km ²	N-S	0.21	0.16	0.31	0.70	-0.01	0.52	0.17	0.13	0.66	0.62	0.11	0.21
Florida Bay	E-W	0.47	0.15	0.64	0.67	0.54	0.52	0.23	0.41	0.37	0.65	0.05	-0.13
392.4 km ²	N-S	0.56	0.12	0.55	0.53	0.2	0.64	0.17	0.61	0.33	0.82	0.33	0.48

Conclusions

- Canal's water ended up in the Gulf and Atlantic Ocean through Taylor River Slough;
- Wind determines the flooded areas in the southern Everglade adjacent to Florida Bay and the muddy banks in Florida Bay;
- Southerly and westerly wind tend to result in more flooded areas, while northerly and easterly wind tend to decrease the flooded areas.

Future works

- Calculate water budget and residence time for sub bays and coastal lakes;
- Analyze seasonal variation for the Everglades' and Florida Bay's hydroperiod;
- Examine salinity and flow pattern for land-to-ocean continuum.
- Try to include groundwater effect

Thanks!

Any Questions?

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